Identification of Dynamical Systems

Project - LTI identification

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1. System description

The device studied throughout this project is a linear system with a static nonlinear feedback loop. It has a single input and a single output and the feedback is known to follow the input-output relation :

$$y(t) = u^3(t) \tag{1.1}$$

The objective will be to identify the linear part of the system and to have a measure of the nonlinearity and noise levels. The knowledge of Eq. 1.1 will not be used, except for a small parenthesis about odd and even non linearities.

The measurement device (referred to as the *PXI setup* in the comments of the Matlab code) allows to send only multisines and their RMS value cannot exceed 1V. Its ADC's have a range that can be chosen and can sample at high frequencies (> 1GHz).

2. Excitation signal

Before making any useful measurement, the parameters of the excitation signal must be chosen to match to the DUT parameters.The rms of the applied signal must indeed be determined to use the full range of the ADC while avoiding saturation of the system. The poles of the system must also be roughly known to determine the band of interest, making sure that the system's dynamics are measured.

2.1 Band of interest

To determine the band of interest, a multisine excited up to a high frequency must be applied (here 1MHz). The spectral resolution of the signal is chosen to be small on order to improve the SNR and not miss any interesting behaviour which could be hidden by noise.

TODO

2.2 Amplitude

TODO

2.3 Transient removal

Because only multisines can be applied, every measurement is done for multiple periods and the first ones are systematically removed until the transient is removed. To check this, the difference between each period is plotted, as shown in Fig. 2.1 and periods are removed until the shown plot is reduced to a flat curve.

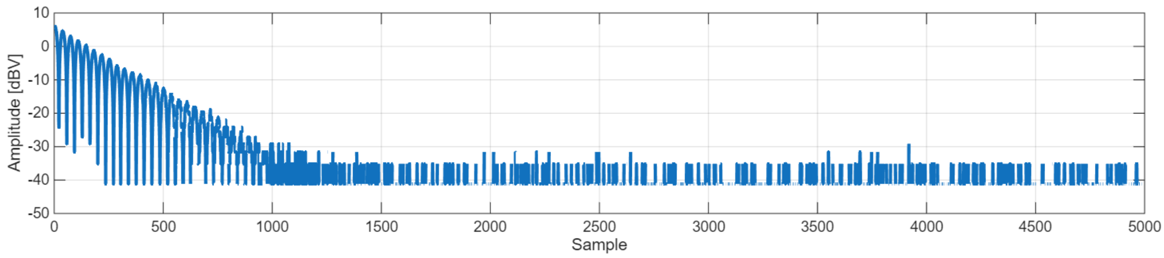

Figure 2.1: Difference between 1st and 2nd periods of the output signal in time domain

3. Nonlinear distortion and noise analysis

Two methods are used to distinguish the nonlinear distortions from the noise. The *fast method* uses a single experiment and allows to split the output spectrum in linear, nonlinear and noise contributions. The *robust method* uses multiple experiments and computes an estimate of the variances of nonlinear distortion and noise together with an non parametric estimate of the FRF.

3.1 Fast method

The excitation signal is an odd multisine with holes. This means that only odd frequencies are excited and some of those odd frequencies are also unexcited in order to have a measure for odd nonlinear distortions. During the excitation generation, 1 odd bin out of 4 is randomly chosen to be not excited.

The FFT of multiple periods are taken, effectively dividing the spectral resolution as $\Delta_f = f_s/N$ with a larger N. The added bins are not linear combinations of input frequencies and therefore only contain noise, under the assumption that the DUT is a *PISPO*[1] system.

The spectrum of the output signal is then divided into 4 categories:

- **Linear**, for bins that were excited.
- **Even distortions**, for bins that are a combination of an even number of excited bins.
- **Odd distortions**, for bins that are a combination of an odd number of excited bins.
- **Noise**, for bins that are added by the multiple periods.

Fig. 3.1 shows that even distortion is close to the noise level and therefore does not impact much the signal. Odd distortion on the other hand is quite high and is close to the level of the linear part near its peak.

This behaviour is explained by Eq. 1.1: because the third power of the output is fed back to the input, it contains sinusoids at the sum of 3 frequencies of the output. As the output only contains odd frequencies, it results in another odd frequency and the main non linearities therefore appears at odd bins.

[1] period in, same period out

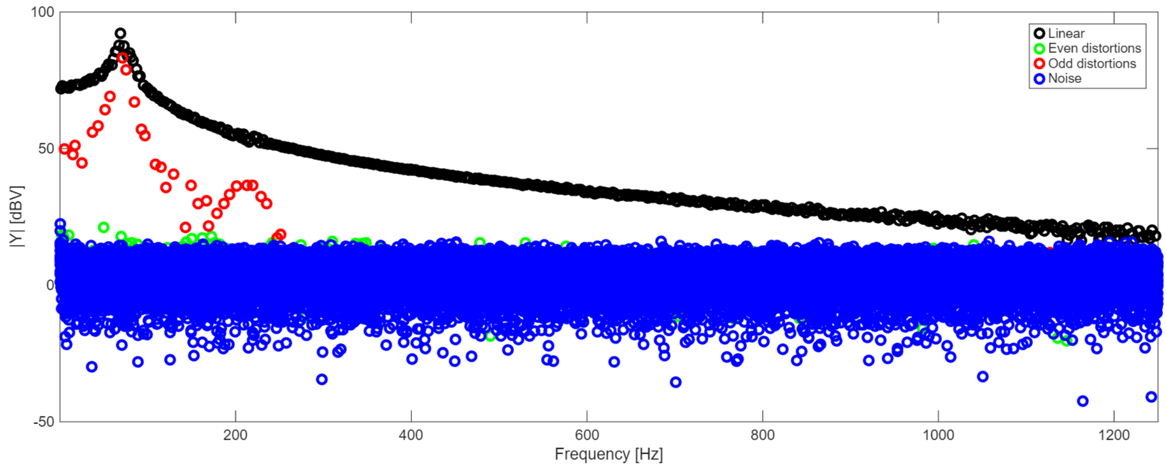

Figure 3.1: Results of the fast method

To validate this explanation, the same experiment has been applied to the system with this time as feedback:

$$y(t) = u^2(t) \tag{3.1}$$

The feedback now excites even frequencies and will therefore have more even distortions than what is shown on Fig. 3.1.

3.2 Robust method

This method relies on computing a FRF for each of the P periods and the M realizations. The variance across the periods for realization m is called $\hat{\sigma}_n^{2[m]}$ and is only due to noise. This is because the non linear contributions are deterministic and depend only on the input signal, which is the same for the whole realization.

$$\hat{G}^{[m,p]} = \frac{Y^{[m,p]}}{U^{[m]}} \tag{3.2}$$

$$\hat{G}^{[m]} = \frac{1}{P}\sum_{p=1}^{P} \hat{G}^{[m,p]} \qquad \hat{\sigma}_n^{2[m]} = \frac{1}{P(P-1)}\sum_{p=1}^{P} \left|\hat{G}^{[m,p]} - \hat{G}^{[m]}\right|^2 \tag{3.3}$$

These noise variances on a single realization $\hat{\sigma}_n^{2[m]}$ are then averaged across the realizations and divided by M because the averaging reduces the variance. This gives the total noise variance σ_n^2.

$$\hat{\sigma}_n^2 = \frac{1}{M^2}\sum_{m=1}^{M} \hat{\sigma}_n^{2[m]} \tag{3.4}$$

The averaged FRF of a single realization $\hat{G}^{[m]}$ doesn't contain noise (at least in expected value). But between realizations, they have different non linear effects as the phases of the excitation signals have changed. The variance between those therefore is due to nonlinear distortion and averaging it out gives a FRF estimate that is noiseless and purely linear, if M and P are large enough to be close to the expected value.

$$\hat{G}_{\mathrm{ML}} = \frac{1}{M}\sum_{m=1}^{M}\hat{G}^{[m]} \qquad\qquad \hat{\sigma}_{\mathrm{ML}} = \frac{1}{M(M-1)}\sum_{m=1}^{M}\left|\hat{G}^{[m]} - \hat{G}_{\mathrm{ML}}\right|^2 \tag{3.5}$$

A full multisine (excited up to 1.25kHz) has been applied to the DUT for $P = 20$ periods and $M = 20$ realizations. The results are shown on Fig. 3.2.

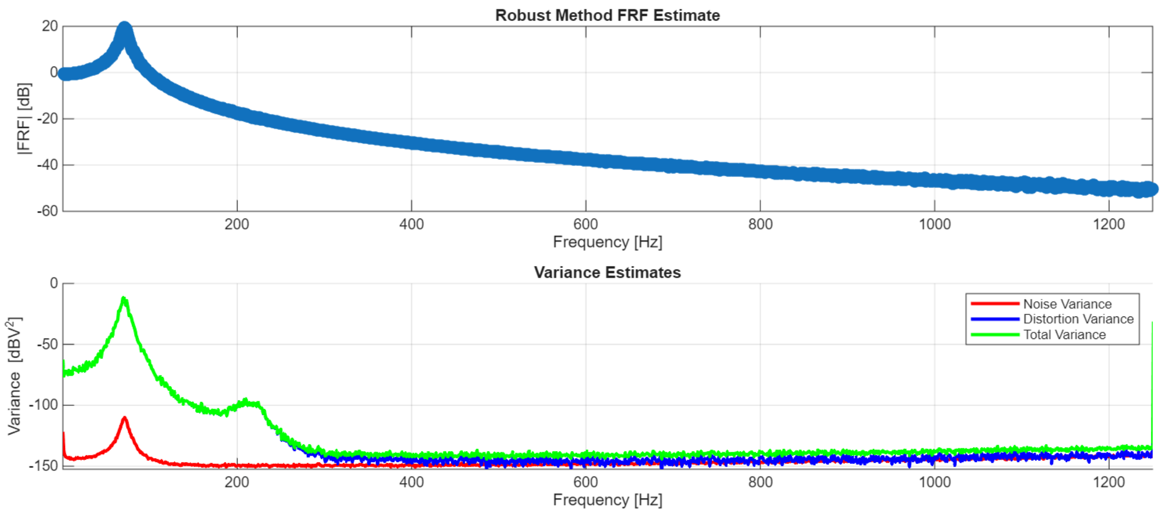

Figure 3.2: Results of the robust method

4. Best linear approximation

A linearized model of the system can be determined to approximate the DUT as an LTI. The best linear approximation, or BLA, is therefore computed as the "best fit". Because nonlinear distortions are dependent on the input signal, the BLA computed here is valid for excitation signals with a similar *Riemann equivalent spectrum*.

4.1 Non parametric model

The result of the robust method shown in Fig. 3.2 is, in fact, an estimated BLA and it already comes with its uncertainty. Because of the successive averaging, it should be able to remove noise and nonlinear distortions which are zero-mean [1].

4.2 Parametric model

Based on the non parametric model of Sec. 4.1, a parametric model can be derived by making a choice for the model. This will be fixed by the order of the numerator and denominator of the transfer function, respectively n_b and n_a such that:

[1] It is only true in expected value so it can only be approached using a discrete number of periods and realizations

$$G(s) = \frac{\sum_{k=0}^{n_b} b_k s^k}{\sum_{k=0}^{n_a} a_k s^k} \tag{4.1}$$

Different estimators are used but first, a few tricks are used for each of them and those are explained here.

Normalization by rms scaling: Each estimator has a regressor matrix (H or J), whose columns often have different orders of magnitude. This results in a poor conditioning and gives numerical errors. To resolve this, a change of coordinates is applied with the scaling matrix S, which scales the columns. The original coordinates H (or J) and θ are replaced by $\tilde{H}$ (or $\tilde{J}$) and $\tilde{\theta}$:

For LLS

$$y = H \times \theta \quad \Rightarrow \quad y = \tilde{H} \times \tilde{\theta} \quad \text{with} \quad \tilde{H} = H \times S \tag{4.2}$$

$$\Leftrightarrow \quad \tilde{\theta} = S^{-1} \times \theta \tag{4.3}$$

$$\Leftrightarrow \quad \theta = S \times \tilde{\theta} \tag{4.4}$$

S has been chosen to be a diagonal matrix with as elements the inverse of the rms of the columns of H, such that it is invertible:

$$S = \operatorname{diag}\left(\frac{1}{\operatorname{rms}(H_{:,1})}, \ldots, \frac{1}{\operatorname{rms}(H_{:,n})}\right) \tag{4.5}$$

For TLS

The exact same thing is applied, except y is now called e and H becomes J.

Use of the non parametric estimate: All the estimators are relying on approaching the following, which arises from the definition of a transfer function:

$$Y(k) = G(j\omega_k) \times U(k) \tag{4.6}$$

The key idea is to notice that

$$U(k) = 1 \quad \forall k \quad \Rightarrow \quad Y(k) = G(j\omega_k) \tag{4.7}$$

Forcing real valued parameters: A real system is described by real coefficients so the imaginary part of θ must be equal to 0

For LLS and TLS (replacing H by J)

$$y = H \times \theta \quad \Rightarrow \quad \Re\{y\} + j\Im\{y\} = \left(\Re\{H\} + j\Im\{H\}\right) \times \left(\Re\{\theta\} + j\Im\{\theta\}\right) \tag{4.8}$$

$$\Leftrightarrow \quad \begin{bmatrix} \Re\{y\} \\ \Im\{y\} \end{bmatrix} = \begin{bmatrix} \Re\{H\} & -\Im\{H\} \\ \Im\{H\} & \Re\{H\} \end{bmatrix} \times \begin{bmatrix} \Re\{\theta\} \\ \Im\{\theta\} \end{bmatrix} \tag{4.9}$$

And because $\Im\{\theta\} = 0$, it becomes:

$$\begin{bmatrix} \Re\{y\} \\ \Im\{y\} \end{bmatrix} = \begin{bmatrix} \Re\{H\} \\ \Im\{H\} \end{bmatrix} \times \Re\{\theta\} \tag{4.10}$$

Thus y and H are replaced by a concatenation of their real and imaginary parts called y_{re} and H_{re}.

For GTLS

As explained later on, the GTLS estimator needs the covariance matrix $C_J = \mathbb{E}\left\{\Delta J^H \Delta J\right\}$, with $J = J_0 + \Delta J$. It must be adapted because J has become J_{re}.

$$\Delta J_{\text{re}} = \begin{bmatrix} \Re\{\Delta J\} \\ \Im\{\Delta J\} \end{bmatrix} \qquad \Rightarrow \qquad C_{J_{\text{re}}} = \mathbb{E}\left\{ \begin{bmatrix} \Re\{\Delta J\}^T & \Im\{\Delta J\}^T \end{bmatrix} \times \begin{bmatrix} \Re\{\Delta J\} \\ \Im\{\Delta J\} \end{bmatrix} \right\} \tag{4.11}$$

$$= \mathbb{E}\left\{\Re\{\Delta J\}^T \times \Re\{\Delta J\}\right\} + \mathbb{E}\left\{\Im\{\Delta J\}^T \times \Im\{\Delta J\}\right\} \tag{4.12}$$

By splitting the real and the imaginary part of ΔJ, C_J can be computed as follows:

$$\Delta J = \Re\{\Delta J\} + j\Im\{\Delta J\} \qquad \Rightarrow \qquad \begin{aligned} C_J = &\ \mathbb{E}\left\{\Re\{\Delta J\}^T \Re\{\Delta J\}\right\} + \mathbb{E}\left\{\Im\{\Delta J\}^T \Im\{\Delta J\}\right\} \\ &+ j\left(\mathbb{E}\left\{\Re\{\Delta J\}^T \Im\{\Delta J\}\right\} - \mathbb{E}\left\{\Im\{\Delta J\}^T \Re\{\Delta J\}\right\}\right) \end{aligned} \tag{4.13}$$

By comparing Eq. 4.12 with Eq. 4.13, it can be seen that:

$$C_{J_{\text{re}}} = \Re\left\{C_J\right\} \tag{4.14}$$